

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this document is to describe the Software Requirements Specification (SRS) for **Campus Bot**, a centralized digital platform designed for college clubs to manage events and for students to register and participate in those events. This document serves as a reference for developers, designers, testers, and stakeholders involved in the development of the system.

1.2 Scope

Campus Bot aims to simplify event management and participation within a college campus. It provides:

- Club-wise event creation and management
- Student registration and participation tracking
- Automated notifications and reminders
- Role-based access for students, club coordinators, and administrators

The system will reduce manual work, avoid miscommunication, and improve student engagement in campus activities.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS** – Software Requirements Specification
- **Admin** – College authority managing the platform
- **Club Coordinator** – Authorized member managing club events
- **Participant** – Student registering for events
- **UI** – User Interface
- **RBAC** – Role-Based Access Control

1.4 Overview

This document covers system features, functional and non-functional requirements, user roles, constraints, and future enhancements.

2. Overall Description

2.1 Product Perspective

Campus Bot is a web-based platform that acts as a single point of access for all college club events. It can be integrated with the college portal or used as a standalone application.

2.2 User Classes and Characteristics

- **Students:** Register, browse events, participate, receive notifications

- **Club Coordinators:** Create and manage events, view registrations
- **Admin:** Approve clubs/events, manage users, generate reports

2.3 Operating Environment

- Frontend: Web browser (Chrome, Edge, Firefox)
- Backend: Cloud or college server
- Database: Relational or NoSQL database
- Devices: Desktop, laptop, tablet, mobile

2.4 Constraints

- Internet connectivity required
- Only registered college users allowed
- Data privacy and security compliance

2.5 Assumptions and Dependencies

- Students have valid college credentials
- Clubs are officially registered
- Email/SMS services are available for notifications

3. System Features and Requirements

3.1 User Authentication & Authorization

Description: Secure login system for all users.

Functional Requirements:

- Users shall be able to register using college email/ID
- System shall support login and logout functionality
- Role-based access control (Student, Coordinator, Admin)

3.2 Club Management

Description: Manage college clubs on the platform.

Functional Requirements:

- Admin shall approve or reject club registrations
- Club coordinators shall update club details
- Students shall view all active clubs

3.3 Event Creation & Management

Description: Enable clubs to create and manage events.

Functional Requirements:

- Coordinators shall create events with details (title, date, venue, rules)
- System shall allow event editing and deletion
- Admin shall approve events before publishing

3.4 Student Event Registration

Description: Allow students to register for events.

Functional Requirements:

- Students shall browse upcoming and past events
- Students shall register or unregister for events
- System shall limit registrations based on capacity

3.5 Notifications & Announcements

Description: Inform users about events and updates.

Functional Requirements:

- System shall send confirmation notifications after registration
- System shall notify students about event updates or cancellations
- System shall send reminders before event date

3.6 Attendance & Participation Tracking

Description: Track student participation.

Functional Requirements:

- Coordinators shall mark attendance
- System shall generate participant lists
- Admin shall view participation analytics

3.7 Feedback & Rating System (Additional Feature)

Description: Collect feedback after events.

Functional Requirements:

- Students shall submit feedback and ratings

- Coordinators shall view feedback reports
- Admin shall analyze event quality

3.8 Certificate Generation (Additional Feature)

Description: Digital certificates for participants and winners.

Functional Requirements:

- System shall auto-generate participation certificates
- Certificates shall include student name, event name, and date
- Students shall download certificates

3.9 Chatbot Assistance – Campus Bot (Core Feature)

Description: AI-based chatbot for guidance.

Functional Requirements:

- Bot shall answer queries about events and clubs
- Bot shall guide students through registration steps
- Bot shall provide deadlines and event details

4. External Interface Requirements

4.1 User Interface

- Simple and responsive UI
- Dashboard for each user role
- Search and filter options for events

4.2 Hardware Interface

- Compatible with standard devices (PC, mobile)

4.3 Software Interface

- Email/SMS gateway for notifications
- Database for storing user and event data

5. Non-Functional Requirements

5.1 Performance

- System should support concurrent users
- Event pages should load within 2 seconds

5.2 Security

- Encrypted passwords
- Secure authentication
- Data privacy for student information

5.3 Usability

- User-friendly navigation
- Minimal training required

5.4 Reliability

- System uptime of at least 99%
- Backup and recovery mechanisms

5.5 Scalability

- Support addition of new clubs and events
- Handle increased user traffic

6. Future Enhancements

- Mobile application
- QR-code based event check-in
- Integration with college ERP/LMS
- AI-based event recommendations
- Resume points / activity credits for students

7. Conclusion

Campus Bot provides an efficient and centralized solution for managing college club events and student participation. With automation, chatbot support, and analytics, the platform enhances campus engagement and reduces administrative workload.